

Exercise 1: Implementing the Singleton Pattern

Code:

```
public class SingletonPatternExample{
    public static void main(String[] args){

        Logger L1 = Logger.getInstance();
        Logger L2 = Logger.getInstance();

        if(L1 == L2){
            System.out.println("Singleton Pattern Works!");
        }else{
            System.out.println("Singleton Pattern is not working!");
        }
    }
}

class Logger{
    private static Logger obj = new Logger();
    private int data=0;

    private Logger(){
        System.out.println("Logger initiated");
    }

    public static Logger getInstance(){
        return obj;
    }
}
```

Output:

Module-1 > Design Patterns and Principles > SingletonPatternExample.java

```
1 public class SingletonPatternExample{
2     public static void main(String[] args){
3         Logger L1 = Logger.getInstance();
4         Logger L2 = Logger.getInstance();
5
6         if(L1 == L2){
7             System.out.println(x: "Singleton Pattern Works!");
8         }else{
9             System.out.println(x: "Singleton Pattern is not working!");
10        }
11    }
12 }
13
14
15 class Logger{
16     private static Logger obj = new Logger();
17     private int data=0;
18
19     private Logger(){
20         System.out.println(x: "Logger initiated");
21     }
22
23     public static Logger getInstance(){
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\adits\Desktop\DeepSkillig\Coding> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\adits\AppData\Roaming\Code\User\workspaceStorage\3385de95c309e825d680cb68353935b\redhat.java\jdt_ws\Coding_613b24a0\bin' 'SingletonPatternExample'
Logger initiated
Singleton Pattern Works!
PS C:\Users\adits\Desktop\DeepSkillig\Coding>
```

powershell
powershell
Run: Singlet...
Run: Singlet...
Run: Singlet...